Front matter

I'm a little tea pot
Short and stout
Here is my handle
Here is my spout
When I get all steamed up
Hear me shout
Tip me over and
Pour me out

Piecewise Functions

- Sometimes we want to build functions that act differently for different inputs. For example, suppose a business charges $10/pizza, but only $5 for orders of six or more. How could we write a function that computes the total price based on the number of pizzas?
- In math, **Piecewise Functions** are functions that can behave one way for part of their Domain, and another way for a different part. In our pizza example, our function would act like $cost(pizzas) = 10 * pizzas$ for anywhere from 1-5 pizzas. But after 5, it acts like $cost(pizzas) = 5 * pizzas$.
- Piecewise functions are divided into "pieces". Each piece is divided into two parts:
  1. How the function should behave
  2. The domain where it behaves that way
- Our programming language can be used to write piecewise functions, too! Just as in math, each piece has two parts:

```
fun cost(pizzas):
  if pizzas < 6: 10 * pizzas
  else if pizzas >= 6: 5 * pizzas
  end
end
```

Piecewise functions are powerful, and let us solve more complex problems. We can use piecewise functions in a video game to add or subtract from a character's x-coordinate, moving it left or right depending on which key was pressed.

Red Shape - Explore

1) Open the Red Shape Starter File, and read through the code you find there. This code contains new programming that you haven't seen yet! Take a moment to list everything you Notice, and then everything you Wonder…

Notice	Wonder

2) What happens if you click "Run" and type `red-shape("ellipse")` ?

3) Add **another example** for "triangle".

4) Add another line of code to the definition, to define what the function should do with the input "triangle".

5) Come up with some new shapes, and add them to the code. Make sure you include examples or you will get an error message!

6) In your own words, describe how *piecewise functions* work in this programming environment.

Word Problem: red-shape

Directions: A friend loves red shapes so we've decided to write a program that makes it easy to generate them. Write a function called `red-shape` which takes in the name of a shape and makes a 20-pixel, solid, red image of the shape.

Contract and Purpose Statement

Every contract has three parts…

red-shape:: *String* -> *Image*

function name *Domain* *Range*

Given a shape name, produce a solid, red, 20-pixel image of the shape.

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

red-shape(*"circle"*) **is** *circle(20, "solid", "red")*

function name *input(s)* *what the function produces*

red-shape(*"triangle"*) **is** *triangle(20, "solid", "red")*

function name *input(s)* *what the function produces*

red-shape(*"rectangle"*) **is** *rectangle(20, 20, "solid", "red")*

function name *input(s)* *what the function produces*

red-shape(*"star"*) **is** *star(20, "solid", "red")*

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

if _____

else if _____

else if _____

else if _____

else: _____

end

end

Decide & Defend - Piecewise Onto Functions

Joy and Marianna have written two different sets of code to accomplish the same goal of helping a caterer direct people with dietary restrictions to a menu item that works for them. Look at the code below.

Joy's Code:

```
fun entree(diet):
  if diet == "none": lasagna
  else if diet == "gluten-free": salmon
  else if diet == "kosher": salmon
  else if diet == "lactose-int": salmon
  else if diet == "nut allergy": lasagna
  else if diet == "vegan": stir-fry
  else if diet == "vegetarian": stir-fry
  else: text("unknown diet", 20, "red")
  end
end
```

Marianna's Code:

```
fun entree(diet):
  if (diet == "none") or (diet == "nut allergy"): lasagna
  else if ((diet == "gluten-free") or (diet == "kosher")) or (diet == "lactose-int"): salmon
  else if (diet == "vegan") or (diet == "vegetarian"): stir-fry
  else: text("unknown diet", 20, "red")
  end
end
```

Whose method do you like better? Why?

_____

_____

_____

_____

Mood Generator - Explore

1) Open the Mood Generator Starter File, and read through the code you find there. This code contains new programming that you haven't seen yet! Take a moment to list everything you Notice, and then everything you Wonder…

Notice	Wonder

2) Add another line of code to the definition for a "mad" mood that produces the *same* emoji as as an "angry" mood.

3) Add **another example** for "laughing", using the appropriate emoji. *(To bring up the emojis on your computer, type* `Cmd-Ctrl-Space` *on a Mac, or* `Windows-Period` *on Windows 10.)*

4) Come up with some new moods, and add them to the code. Make sure you include examples!

5) In your own words, describe how *piecewise functions* work in this programming environment.

_____

_____

_____

_____

Word Problem: Mood Generator

NOTE: This file uses emojis. Even though emojis look like images, they are actually characters in a string! They can be accessed from your keyboard, just like any other character.

Directions: They say a picture is worth a thousand words. Write a function `mood` that translates moods into emojis so that we can "see" what someone is feeling.

Contract and Purpose Statement

Every contract has three parts...

mood:: *String* -> *String*

function name *Domain* *Range*

Consumes a mood and produces the emoji for that mood.

what does the function do?

Examples

Write some examples, then circle and label what changes...

examples:

mood("*happy*") **is** "😀"

function name *input(s)* *what the function produces*

mood("*sad*") **is** "😥"

function name *input(s)* *what the function produces*

mood("*angry*") **is** "😡"

function name *input(s)* *what the function produces*

mood("*sick*") **is** "🤮"

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values...

fun _____ (_____):

function name *variable(s)*

if _____

else if _____

else if _____

else if _____

else: _____

end

end

Welcome to Alice's Restaurant!

Alice has hired you to improve some code used at the restaurant. The code we'll be improving on is shown below.

Read through the code line-by-line with your partner before writing down your observations in the tables below.

```
# cost :: String -> Number
# given a item, produce the cost of that item
```

```
fun cost(item):
  if item == "hamburger": 6.0
  else if item == "onion rings": 3.5
  else if item == "fried tofu": 5.25
  else if item == "pie": 2.25
  else: "Sorry, that's not on the menu!"
  end
end
```

1) **I notice...**	2) **I wonder...**
3) **Familiar things I see in the code**	4) **Unfamiliar things I see in the code**

Alice's Restaurant - Explore

Alice's code has some new elements we haven't seen before, so let's experiment a bit to figure out how it works! **Open the [Alice's Restaurant Starter File](), click "Run", and try using the `cost` function in the Interactions window.**

1) What does `cost("hamburger")` evaluate to? _____

2) What does `cost("pie")` evaluate to? _____

3) What if you ask for `cost("fries")`? _____

4) Explain what the function is doing in your own words. _____

_____

_____

5) What is the function's name? Domain? _____

6) What is the name of its variable? _____

7) Alice says onion rings have gone up to $3.75. Change the `cost` function to reflect this.

8) Try adding menu items of your own. What's your favorite?

9) For an unknown food item, the function produces the String `"That's not on the menu!"` Is this a problem? Why or why not?

_____

_____

10) Suppose Alice wants to calculate the price of a hamburger, *including a 5% sales tax*. Draw a Circle of Evaluation for the expression below.

Word Problem: Alice's Restaurant

Directions: Alice's Restaurant has hired you as a programmer. They offer the following menu items: hamburger ($6.00), onion rings ($3.50), fried tofu ($5.25) and pie ($2.25). Write a function called `Alice's Restaurant` which takes in the name of a menu item and outputs the price of that item.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____

function name *Domain* *Range*

_____

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):

function name *variable(s)*

if _____

else if _____

else if _____

else if _____

else: _____

end

end

Back matter

Some say the world will end in fire
Some say in ice
From what I know of desire
I hold with those who favor fire
But if it had to perish twice
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice

—R. Frost